

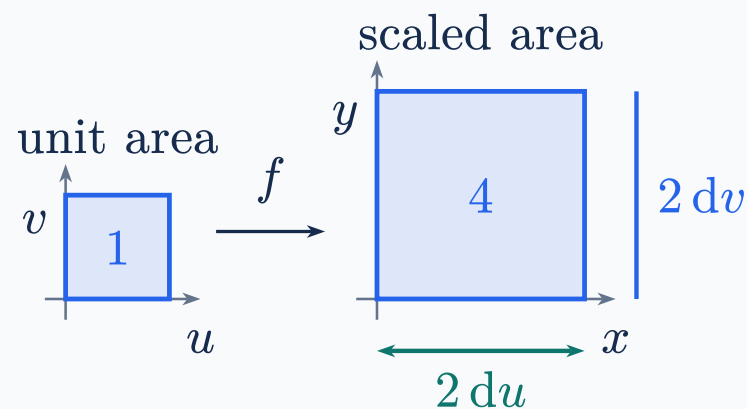
Change of variables

The Jacobian measures how a tiny coordinate cell changes area or volume.

$$\int_{f(U)} g(\mathbf{x}) \, d\mathbf{x} = \int_U g(f(\mathbf{u})) \left| \det J_{f(\mathbf{u})} \right| \, d\mathbf{u}$$

Linear scaling

$$x = 2u, \quad y = 2v$$



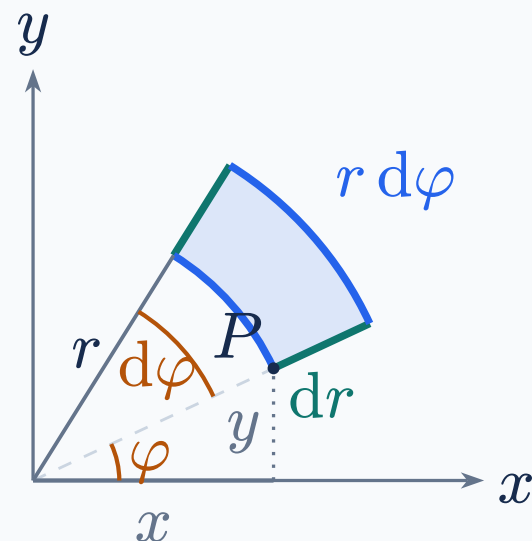
$$2 \times 2 = |\det J_f| = 4$$

$$dA = 4 \, du \, dv$$

Each side doubles, so area grows by four. Reflecting one axis gives $\det J_f = -4$; the area factor stays 4.

Polar coordinates

$$x = r \cos \varphi, \quad y = r \sin \varphi$$



$$1 \times r = |\det J_f| = r$$

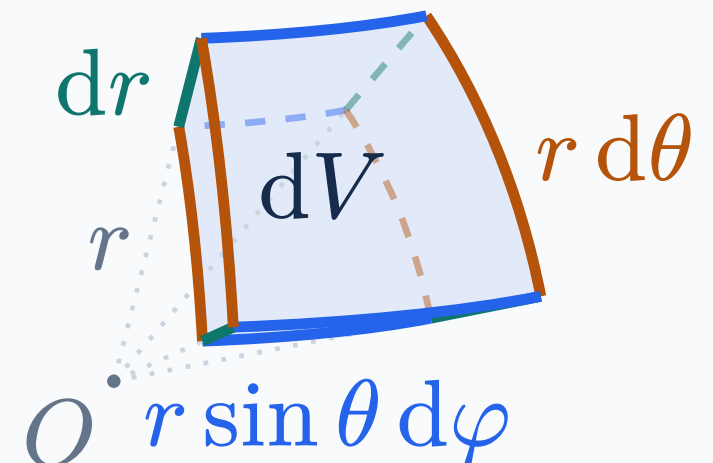
$$dA = r \, dr \, d\varphi$$

The angular side is arc length $r \, d\varphi$. Multiply it by the radial thickness dr to get the area element.

Spherical coordinates

$$x = r \sin \theta \cos \varphi$$

$$y = r \sin \theta \sin \varphi, \quad z = r \cos \theta$$



$$1 \times r \times r \sin \theta = r^2 \sin \theta$$

$$dV = r^2 \sin \theta \, dr \, d\theta \, d\varphi$$

Meridians have radius r ; latitude circles have radius $r \sin \theta$. Their arc lengths supply the two angular factors.

Use the absolute determinant. Orientation can flip; an area or volume measure stays nonnegative. The map f is one-to-one on U .

Angles are in radians; θ is measured from $+z$. Cells are enlarged for clarity; labeled lengths describe the infinitesimal limit.